4.4.4 A has linearly dependent columns

Let us now consider the case where $A \in \mathbb{C}^{m \times n}$ has rank $r \leq n$. In other words, it has r linearly independent columns. Let $p \in \mathbb{R}^n$ be a permutation vector, by which we mean a permutation of the vector

$$\begin{pmatrix} 0 \\ 1 \\ \vdots \\ n-1 \end{pmatrix}$$

And $P(p)$ be the matrix that, when applied to a vector $x \in \mathbb{C}^n$ permutes the entries of x according to the vector p:

$$P(p)x = \underbrace{\begin{pmatrix} e_{\pi_0}^T \\ e_{\pi_1}^T \\ \vdots \\ e_{\pi_{n-1}}^T \end{pmatrix}}_{P(p)} x = \begin{pmatrix} e_{\pi_0}^T x \\ e_{\pi_1}^T x \\ \vdots \\ e_{\pi_{n-1}}^T x \end{pmatrix} = \begin{pmatrix} \chi_{\pi_0} \\ \chi_{\pi_1} \\ \vdots \\ \chi_{\pi_{n-1}} \end{pmatrix}.$$

where e_j equals the columns of $I \in \mathbb{R}^{n \times n}$ indexed with j (and hence the standard basis vector indexed with j).

If we apply $P(p)^T$ to $A \in \mathbb{C}^{m \times n}$ from the right, we get

$$\begin{array}{l} AP(p)^T \\ \quad = \quad < \text{definition of } P(p) > \\ A \begin{pmatrix} e_{\pi_0}^T \\ \vdots \\ e_{\pi_{n-1}}^T \end{pmatrix}^T \\ \quad = \quad < \text{transpose} > \\ A \left(\begin{array}{c|c|c} e_{\pi_0} & \cdots & e_{\pi_{n-1}} \end{array} \right) \\ \quad = \quad < \text{matrix multiplication by columns} > \\ \left(\begin{array}{c|c|c} Ae_{\pi_0} & \cdots & Ae_{\pi_{n-1}} \end{array} \right) \\ \quad = \quad < Be_j = b_j > \\ \left(\begin{array}{c|c|c} a_{\pi_0} & \cdots & a_{\pi_{n-1}} \end{array} \right). \end{array}$$

In other words, applying the transpose of the permutation matrix to A from the right permutes its columns as indicated by the permutation vector p.

The discussion about permutation matrices gives us the ability to rearrange the columns of A so that the first $r = \text{rank}(A)$ columns are linearly independent.

Theorem 4.4.4.1. *Assume $A \in \mathbb{C}^{m \times n}$ and that $r = \text{rank}(A)$. Then there exists a permutation vector $p \in \mathbb{R}^n$, orthonormal matrix $Q_L \in \mathbb{C}^{m \times r}$, upper triangular matrix $R_{TL} \in \mathbb{C}^{r \times r}$, and $R_{TR} \in \mathbb{C}^{r \times (n-r)}$ such that*

$$AP(p)^T = Q_L \left(\begin{array}{c|c} R_{TL} & R_{TR} \end{array} \right).$$

Proof.

Let us examine how this last theorem can help us solve the LLS

$$\text{Find } \widehat{x} \in \mathbb{C}^n \text{ such that } \|b - A\widehat{x}\|_2 = \min_{x \in \mathbb{C}^n} \|b - Ax\|_2$$

when $\text{rank}(A) \leq n$:

$$\begin{array}{l} \min_{x \in \mathbb{C}^n} \|b - Ax\|_2 \\ \quad = \quad < P(p)^T P(p) = I > \\ \min_{x \in \mathbb{C}^n} \|b - AP(p)^T P(p) x\|_2 \\ \quad = \quad < AP(p)^T = Q_L \left(\begin{array}{c|c} R_{TL} & R_{TR} \end{array} \right) > \\ \min_{x \in \mathbb{C}^n} \|b - Q_L \underbrace{\left(\begin{array}{c|c} R_{TL} & R_{TR} \end{array} \right) P(p) x}_{w} \|_2 \\ \quad = \quad < \text{substitute } w = \left(\begin{array}{c|c} R_{TL} & R_{TR} \end{array} \right) P(p) x > \\ \min_{w \in \mathbb{C}^r} \|b - Q_L w\|_2 \end{array}$$

which is minimized when $w = Q_L^H b$. Thus, we are looking for vector $\widehat{x}$ such that

$$\left(\begin{array}{c|c} R_{TL} & R_{TR} \end{array} \right) P(p) \widehat{x} = Q_L^H b.$$

Substituting

$$z = \left(\begin{array}{c} z_T \\ \hline z_B \end{array} \right)$$

for $P(p)\widehat{x}$ we find that

$$\left(\begin{array}{c|c} R_{TL} & R_{TR} \end{array} \right) \left(\begin{array}{c} z_T \\ \hline z_B \end{array} \right) = Q_L^H b.$$

Now, we can pick $z_B \in \mathbb{C}^{n-r}$ to be an arbitrary vector, and determine a corresponding z_T by solving

$$R_{TL} z_T = Q_L^H b - R_{TR} z_B.$$

A convenient choice is $z_B = 0$ so that z_T solves

$$R_{TL} z_T = Q_L^H b.$$

Regardless of choice of z_B, the solution $\widehat{x}$ is given by

$$\widehat{x} = P(p)^T \left(\begin{array}{c} R_{TL}^{-1}(Q_L^H b - R_{TR} z_B) \\ \hline z_B \end{array} \right).$$

(a permutation of vector z.) This defines an infinite number of solutions if $\mathrm{rank}(A) < n$.

The problem is that we don't know which columns are linearly independent in advance. In enrichments in Subsection 4.5.1 and Subsection 4.5.2, rank-revealing QR factorization algorithms are discussed that overcome this problem.